Conformal Coverage Fails Under Grouped and Temporal Shift but Uncertainty Abstention Repairs Selective Risk: A Preregistered Tabular Audit

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Abstract

This paper provides a deployment-facing validation protocol and decision rule for deciding whether a keyed tabular model's reliability claim survives grouped or temporal shift, and whether uncertainty-ranked abstention is a defensible selective-risk fallback. Across 14 public OpenML datasets with a defensible entity/time key, four open configurations in three architecture families (XGBoost, LightGBM, and the tabular foundation models TabICLv2 and TabDPT), an exploratory Folktables/ACS arm, and a held-out preregistered TableShift evaluation, plain split-conformal falls below its 0.90 target under grouped and temporal splits (median worst-group coverage ≈ 0.81), and no tested distribution-free remedy restores it at the majority level (H1 REJECT). Ranking the same test items by an uncertainty score and abstaining on the highest-risk ones beats random deferral on 13 of 14 exploratory datasets and matches a tuned learned selector on the held-out TableShift evaluation (H2 CONFIRM), though both remain far below an oracle ceiling (median repair ratio 0.34 vs. 0.75). Our preregistered Stage-2 gate evaluated to KILL

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on the degradation count due to split-realization fragility, so that count is reported as a split-realization distribution, not a confirmed majority finding. We recommend plain split-conformal plus uncertainty abstention for grouped tabular shift, scoped strictly to selective reliability rather than a coverage guarantee.
Keywords: Selective prediction, Conformal prediction, Distribution shift, Tabular foundation models, Reliability evaluation, Reproducibility

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1. Introduction

Tabular models are almost universally evaluated with random hold-out splits, yet the shifts that matter in deployment are almost never random: the fraud-detection model trained on known accounts is applied to a new merchant; the demand model trained on historical months must forecast an upcoming quarter; the healthcare risk model trained at established sites is deployed at a new facility. The split that governs reliability at inference is a grouped or temporal split, not a random one. If random-split validation systematically overstates reliability under such shifts, then published performance figures are misleading and practitioners are blind to actual deployment risk—yet this gap is rarely closed with controlled experiments on standard benchmarks. Dataset shift is a classical problem [1]; tabular benchmarks now curate real grouped and temporal shifts and score predictive performance [2, 3, 4], while split conformal supplies a finite-sample coverage guarantee only under exchangeability [5, 6]. What remains missing is a split-conditioned *reliability* audit: does that guarantee, and the associated risk–coverage curve [7, 8], survive the deployment split, and if not, does a free uncertainty score repair selective risk?

We ask three empirical questions about the primary phenomenon, plus one negative-control question about a candidate remedy. (i) Do grouped/temporal splits cause a measurable reliability collapse—degraded risk–coverage and conformal under-coverage? (ii) Does uncertainty-ranked abstention improve selective reliability over random deferral? (iii) Is the effect model-agnostic, reproducing across independent model families? As a negative control on a natural remedy candidate, we additionally ask (iv) whether an *uncertainty-taxonomy* (confidence-conditional) Mondrian conformal predictor helps; it does not (section 3), and we report that compact negative result and its own key-dependent explanation as a control rather than a co-equal finding. (We use the uncertainty-decile Mondrian because a group-keyed taxonomy provably degenerates to split conformal under leave-group-out, where the grouping key is absent from calibration.) Uncertainty-ranked abstention is an instance of selective classification [7, 8]—the learning-theoretic descendant of the classical reject option, from Chow's [9] error–reject tradeoff through statistically grounded classification with a reject option [10] to margin-based learning with rejection [11]—and selective mechanisms are known to interact non-trivially with subgroup structure [12], with reject options themselves able to shape subgroup disparities [13]—a point our Folktables/ACS generalization arm returns to; the conformal machinery we audit is standard split (inductive) conformal [14, 5, 6]. We answer these questions on 14 public, keyed OpenML datasets with defensible entity- or time-split keys [15], using two open gradient-boosted models (XGBoost [16], LightGBM [17]) and two open-weights tabular foundation models (TabICLv2 [18], TabDPT [19]), all token-free.

We therefore make two contributions, with the audit serving as evidence for a reusable decision procedure rather than as the contribution itself:

1. **Transferable protocol and decision rule.** We package a split-type $\times$ reliability $\times$ selective-abstention validation protocol that a practitioner can apply to any keyed tabular model: compare random with leave-group-out or rolling-origin evaluation, test whether split-conformal coverage survives the deployment split, and, when it does not, use the model-native uncertainty score to rank abstention as a selective-risk decision rule. The rule is explicitly scoped to retained-set selective reliability, not to restoring a coverage guarantee.
2. **Supporting empirical verdicts.** We provide preregistered and held-out evidence for that procedure: grouped/temporal validation exposes the coverage failure, while free uncertainty-ranked abstention matches a budget-matched tuned selector across the generalization test; the same audit also records the negative remedy results and the underpowered, knife-edge grouped-gap caveat.

Two results are held to a preregistered, held-out standard, and both hold. First, uncertainty-ranked abstention beats random deferral on 13 of 14 datasets for each of four open, token-free configurations spanning three architecture families—two gradient-boosted models (XGBoost, LightGBM), which behave near-identically, plus two distinct tabular-FM architectures (TabICLv2, TabDPT)—with and without the group key; in the GBM arms it further holds across default and tuned hyperparameters (12–13/14) and in every one of K=20–30 repeated split draws. Beating random deferral is, on its own, a low bar: any score whose confidence is even weakly informative about error will clear it, so the count by itself is not the substantive claim. What is substantive, and what the held-out, preregistered TableShift evaluation confirms without exception at every split-repeat, is that the free uncertainty score ties-or-beats a budget-matched tuned selector (H2 CONFIRM; section 3) on data that could not have generated the hypothesis—so the advantage holds under grouped and temporal shift, is model-agnostic across three architecture families including two foundation models that do not auto-repair the underlying gap, and costs nothing over the model's own confidence—though both the free and tuned scores remain well below an oracle ceiling (section 3: median repair ratio 0.339 free vs. 0.745 oracle), leaving room the field has not yet claimed. Second, the same generalization test REJECTs coverage restoration (H1): no tested distribution-free remedy restores group-conditional coverage on a majority of evaluable tasks, corroborating rather than contradicting the diagnosis, since we never recommend those remedies as a coverage fix. What abstention repairs is the retained-set selective risk—a measured median $\sim$33% reduction relative to random deferral (section 3)—not the coverage gap itself: we do not claim to close a measured fraction of the random$\to$grouped coverage gap (which we cannot, lacking persisted matched-coverage risk–coverage curves), to beat any baseline stronger than random deferral, or that abstention restores the coverage that degrades under grouped/temporal shift; these are different quantities, kept distinct throughout.

The degradation these two confirmed results are read against is real in direction—grouped splits degrade risk–coverage across all four configurations and push split-conformal coverage below the 0.90 nominal target on the temporal splits, including two tabular FMs that do not auto-correct the gap—but its dataset-count is not itself a robust, preregistered claim, rather than let the positive results above imply otherwise: our conjunctive Stage-2 gate on that count evaluated to KILL, the confirmatory/exploratory partition we draw throughout is accordingly a *post-hoc reinterpretation* adopted after that failure, and section 5 gives the full rigor check—why the gap count is realization-, hyperparameter-, and key-fragile, and why we read the degradation as a direction rather than a majority-count finding. The uncertainty-taxonomy Mondrian add-on is a further negative control, not a co-equal finding (section 3): it is not a free win, and its calibration-size explanation is itself fragile—split-fragile (significant on only 4 of 20 split re-draws) and key-dependent (it clears the preregistered bar with the key as a feature but not without it)—an instance of the same sensitivity we document for the headline count, now applied reflexively to a candidate mechanistic explanation, which does not survive it.

Related work.. A 2026 wave of work independently corroborates that tabular-FM accuracy does not buy reliability. De Melo Costa et al. [20] report that tabular FMs beat GBDTs on AUC yet have worse conditional conformal coverage on 112 TALENT datasets—the same "high performance, low reliability" headline—but do not condition on split protocol (random vs. grouped/temporal), report risk–coverage/AURC, or test an abstention repair. The BeyondArena benchmark [21] evaluates tabular FMs across IID/temporal/grouped splits on 142 datasets and finds FMs lead only on small IID data, but reports accuracy/ranking only—no calibration, conformal coverage, or selective prediction. Cheng et al. [22] show that TabPFN v2 [23] degrades under distribution shift, and Helli et al. [24] propose SCM priors to improve its temporal-shift calibration—the "FM auto-repairs shift" counter-hypothesis that our grouped-split arms refute. Two further 2026 entries study TabPFN uncertainty through decomposition [25] and proper scoring rules [26], but neither conditions on split protocol. A parallel line builds tabular out-of-distribution generalization benchmarks—TableShift [2], WhyShift [3], Wild-Tab [4], and the living TabArena benchmark [27]—that curate real distribution shifts and measure accuracy/robustness; we complement them by conditioning the same split types on reliability (risk–coverage and conformal coverage) and adding an abstention analysis, rather than reporting predictive performance alone. The preregistered tabular-foundation-model uncertainty benchmark [28] occupies adjacent territory but evaluates uncertainty across its benchmark setting; it does not provide our keyed grouped/temporal validation protocol, TableShift-class generalization test, or abstention/remedy ladder. A closely related 2025–26 line develops conformal predictors that restore coverage under (sub)population shift—conformal prediction adaptive to unknown subpopulation shifts [29] and audited conformal prediction under unknown distribution shift [30]. Zhou et al. provide a generic classification-focused audit with group-conditional guarantees; our object is the split-conditioned reliability diagnostic plus a selective-abstention analysis, not a new coverage-restoring estimator. We now evaluate the two standard distribution-free remedies (weighted / covariate-shift conformal and inferred-group Mondrian) in an exploratory coverage-remedy audit (section 3), finding that neither robustly restores group-conditional coverage under leave-group-out, while these stronger subpopulation- and audited-CP estimators remain untested here and an open follow-up rather than a competitor. Two threads in this journal's own recent literature bracket that object: Campagner et al. [31] establish the isomorphisms between conformal prediction and three-way (cautious) decision-making, and He et al. [32] couple conformal and selective classification through a conjunction-subspaces test—placing the conformal-plus-abstention pairing we audit squarely in scope—while reject-option methods for tabular decision pipelines, such as credit-scoring reject inference [33], relevance-aggregation interpretability for tabular networks [34], and streaming conformal prediction for evolving data [35], map the surrounding tabular-reliability terrain. On the coverage side, adaptive conformal inference retunes the miscoverage level online to track a drifting distribution [36], and the reach of distribution-free guarantees toward the group-conditional coverage our leave-group-out split voids is delimited by conditional-validity analyses of inductive conformal predictors [37] and by the jackknife+ [38]. For the tabular substrate itself, controlled comparisons keep finding gradient-boosted trees competitive with or ahead of deep networks on typical tabular data [39, 40]—motivating our GBM-anchored roster—and convex-space oversampling has been proposed for the small, imbalanced tables where calibration data are scarcest [41]. Recent surveys of uncertainty quantification in deep models [42, 43] situate selective reliability within this broader reliability agenda. Against this backdrop, our contribution is a single, precisely scoped one, and we state its defense against a near-tautological reading explicitly: we package the split-type $\times$ reliability $\times$ selective-abstention object with preregistered nulls and evaluate it across gradient-boosted models and two tabular foundation models, showing that one uncertainty-abstention recipe substantially improves selective reliability model-agnostically. Beating random deferral alone would be nearly definitional—any weakly informative score clears that bar—so the primary claims are that the advantage holds under grouped and temporal shift, that no tabular foundation model auto-repairs it, and that the free score matches a budget-matched tuned selector at no extra cost, on both the exploratory suite and a held-out, preregistered TableShift evaluation (section 3), while we scope which counts are realization-robust. We do not claim the four configurations are four independent confirmations—the two GBMs are strongly correlated (section 3), so they count as one architecture family plus two FM architectures. Methodologically, this is an empirical reliability/reproducibility study: we root-cause a cross-run irreproducibility to a shared random-number stream (section 5) and replace single-split counts with repeated-draw percentile CIs, treating that discipline as a first-class result rather than an appendix. The conformal machinery we audit is standard split conformal [5, 6]; abstention is selective classification [7, 8], which interacts with subgroup structure [12]. Section 4 returns to questions (i)–(iv) and to the preregistered pair H1/H2, using the same tables and figures rather than a new story. Figure 1 summarizes the audit design used throughout: the frozen predictor and split-conformal wrapper are held fixed while only the split protocol and the coverage remedy vary, with the exploratory remedy audit quarantined from the generalization TableShift arm.

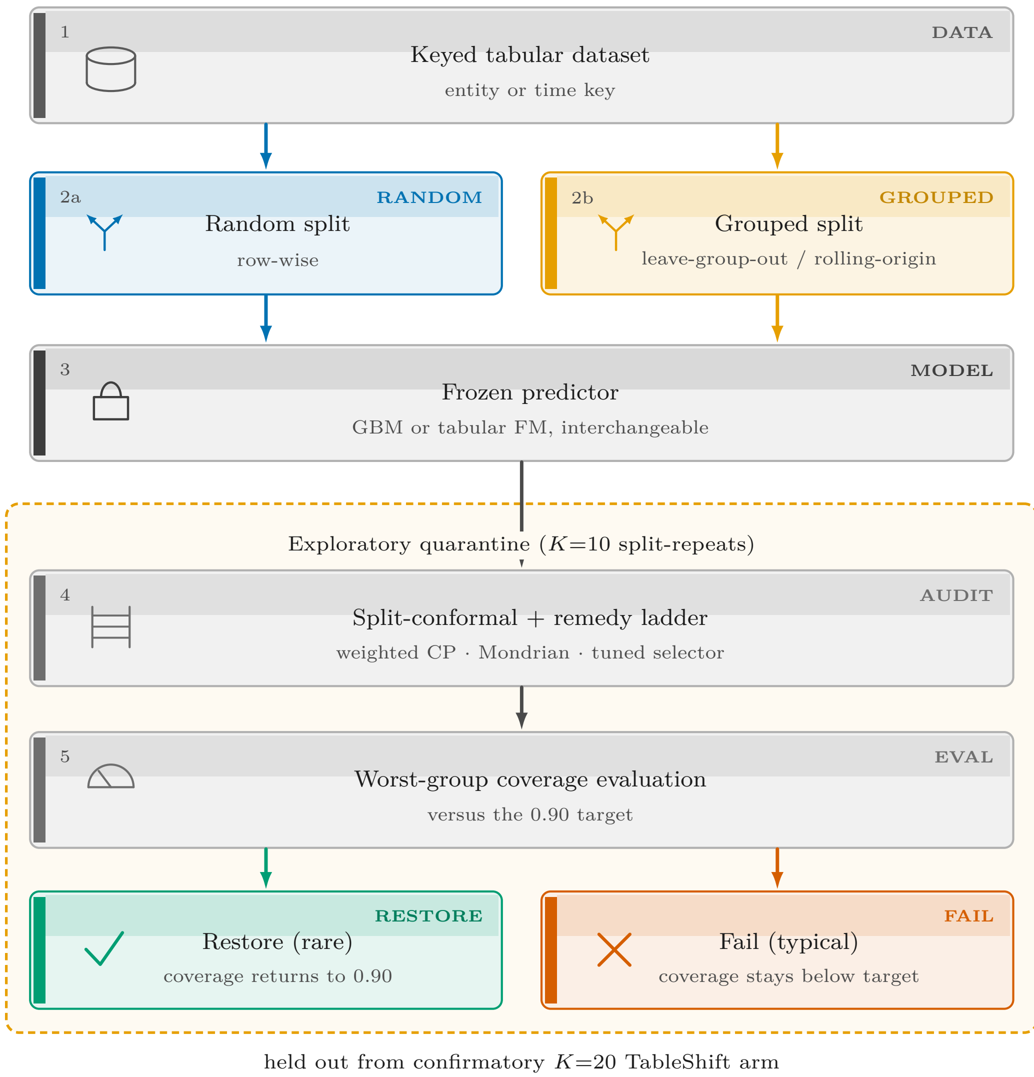

Figure 1: Overview of the reliability-collapse audit design. A keyed tabular dataset (entity or time key) is split on a blue random protocol or a yellow grouped/time protocol (leave-group-out or rolling-origin). A frozen predictor (GBM or tabular FM, interchangeable) is wrapped in split-conformal prediction. A remedy ladder (weighted CP, Mondrian variants, tuned selector) is evaluated by worst-group coverage, yielding Restore (green; coverage returns to the 0.90 target, rare) or Fail (vermillion; coverage remains below target, typical). The remedy audit is exploratory (K=10 split-repeats), held out from the confirmatory K=20 TableShift generalization arm (no data overlap).

2. Data & Methods

Data.. 14 OpenML datasets with a defensible entity key (e.g. Adult/native-country, NYC taxi/location, SF police/district) or time key (e.g. Electricity, Airline delays); selected by a 3-pass census (name → true-cardinality → hand-vetted) from the public OpenML catalog [15]. This suite is deliberately keyed by construction rather than broad: the object of the audit is the reliability gap under a defensible deployment shift, which requires each dataset to carry a genuine entity/time key along which a leave-group-out / rolling-origin split is a real distributional change—most tabular benchmark datasets have none, and an arbitrary-column split is not a shift, so breadth and key-defensibility trade off directly. The 3-pass census reduced 145 candidates to 23 by name, 20 by true cardinality, and 14 hand-vetted with a reviewer-defensible key (frozen after manual review); breadth is supplied separately by the 30-dataset arbitrary-key extension (section 3), kept out of the headline for exactly that reason.

Models.. XGBoost [16] and LightGBM [17] with quantile-interval uncertainty (regression, whose conformalized form is the basis of conformalized quantile regression [44]) / $1 - \max p$ (classification), evaluated both at default hyperparameters and under a leakage-safe 12-trial randomized hyperparameter search (tuned only on the random-split training fold; the grouped test and calibration folds are never seen during tuning).

Foundation-model arms.. On all 14 keyed datasets we additionally run two open-weights, token-free tabular foundation models: TabICLv2 [18] (ensemble of 16, GPU) and TabDPT [19] (8 ensemble members, GPU), each under the same split-conformal + abstention protocol; the same uncertainty score ($1-\max p$ for classification; quantile width for regression) drives the abstention ranking. The TabICLv2 arm is run with the group key excluded and included (a leakage ablation) and against an RNG-reconciled Stage-1 GBM reference; a supplementary in-context scaling sweep (ensemble size $\in \{1, 4, 8, 16\}\times$ context length $\in \{200, 500, 1000, 2000\}$ on four datasets) and a 30-dataset breadth extension (the 14 keyed datasets plus 16 additional OpenML datasets grouped on an arbitrary categorical feature) are reported in section 3 as robustness checks, not as the headline.

Protocol.. For each dataset and model we compute the risk–coverage curve and its area (AURC), together with split-conformal coverage ($\alpha = 0.10$; target 0.90) under a RANDOM split vs. a GROUPED (leave-group-out) / rolling-origin split, with 400-bootstrap CIs [5, 6]. Write $u(x)$ for the model-native uncertainty that ranks abstention and, via the associated nonconformity $s(x, y)$, builds the conformal set: $u(x) = 1 - \max_y \hat{p}_y(x)$ for classification and the predicted quantile-interval width for regression [44, 45]. Let n be the calibration-fold size, $s_i = s(X_i, Y_i)$ the i-th calibration nonconformity, and n_{te} the test-fold size. Split conformal uses the finite-sample calibration quantile of the held-out scores $\{s_i\}_{i=1}^n$,

$$\hat{q} = \text{Quantile}\Big(\{s_i\}_{i=1}^n,\ \tfrac{\lceil (n+1)(1-\alpha)\rceil}{n}\Big), \tag{1}$$

$$\hat{C}(x) = \big\{y : s(x, y) \le \hat{q}\big\}. \tag{2}$$

Let G be the grouping (or time) key and g a group. Write cov_g for the empirical coverage on test points with $G = g$, and $\text{cov}_{\text{best}} = \max_g \text{cov}_g$.

Realized marginal coverage, worst-group coverage, and the nonnegative best–worst group gap are

$$\begin{aligned} \mathrm{cov} &= \frac{1}{n_{\mathrm{te}}} \sum_{j=1}^{n_{\mathrm{te}}} \mathbf{1}\big\{Y_j \in \hat{C}(X_j)\big\}, \\ \mathrm{cov}_{\mathrm{worst}} &= \min_g \mathrm{cov}_g, \qquad \Delta_{\mathrm{grp}} = \mathrm{cov}_{\mathrm{best}} - \mathrm{cov}_{\mathrm{worst}}. \end{aligned} \tag{3}$$

with nominal level $1-\alpha = 0.90$; the undercover flag used in tables is $\mathrm{cov} < 0.88$ (a 0.02 slack). An uncertainty-taxonomy Mondrian [46] predictor replaces the single quantile by one quantile $\hat{q}_b$ per decile b of calibration uncertainty, with $b(x)$ the decile bin of $u(x)$,

$$\hat{C}_{\mathrm{Mondrian}}(x) = \big\{y : s(x, y) \le \hat{q}_{b(x)}\big\}. \tag{4}$$

We use this confidence-conditional taxonomy rather than a group-keyed one because, under leave-group-out, the grouping key is absent from the calibration fold, so a group-keyed Mondrian provably degenerates to split conformal (we verified 0/28 deviation empirically). We then run a calibration-size diagnostic relating per-stratum calibration fold size to the Mondrian coverage improvement $\mathrm{cov}_{\mathrm{Mondrian}} - \mathrm{cov}$. To characterize how sensitive the headline counts are to the particular split realization, we additionally repeat the GBM audit over K=20–30 independent grouped/rolling-origin split draws and report percentile CIs on the grouped-gap and repair counts (predeclared before the runs). Figures are generated from the result records.

Selective risk, AURC, and the repair ratio.. Ranking test points by increasing $u(x)$ and retaining the most confident fraction c yields the risk–coverage curve $R(c)$ (mean loss on the retained set), the standard selective-classification summary [7, 8]. The empirical AURC is the right-endpoint Riemann sum

$$\mathrm{AURC} = \sum_{k=1}^{n_{\mathrm{te}}} (c_k - c_{k-1})\, R(c_k), \qquad c_0 = 0, \tag{5}$$

with c_k the retained fraction after k acceptances (lower is better). Uniformly random deferral has constant selective risk equal to the pooled mean loss $\bar{\ell}$, hence $\mathrm{AURC}_{\mathrm{rand}} = \bar{\ell}$ in closed form. The repair ratio reported in Results is the fraction of that random-deferral AURC removed by uncertainty ranking,

$$\rho = \frac{\mathrm{AURC}_{\mathrm{rand}} - \mathrm{AURC}_u}{\mathrm{AURC}_{\mathrm{rand}}}, \tag{6}$$

so $\rho > 0$ means abstention beats random deferral and $\rho = 0$ is a tie. Weighted / covariate-shift conformal [47] replaces (1) by the likelihood-ratio-weighted quantile

$$\hat{q}_w(x) = \inf\Big\{t : \sum_{i=1}^{n} \tilde{w}_i(x)\, \mathbf{1}\{s_i \le t\} + \tilde{w}_{n+1}(x) \ge 1 - \alpha\Big\}, \tag{7}$$

with $\tilde{w}_i(x) = w(X_i)/(\sum_{j=1}^{n} w(X_j) + w(x))$ and $w(x)$ an estimate of the inter-group density ratio $dP_{\mathrm{test}}/dP_{\mathrm{cal}}$; inferred-group Mondrian uses (4) with $b(x)$ a covariate-space cluster (the split key dropped). Neither construction restores a coverage guarantee without extra assumptions, which is the content of the H1 audit.

Validity of the guarantee.. Split-conformal coverage

$$\Pr\big(Y_{n+1} \in \hat{C}(X_{n+1})\big) \ge 1 - \alpha \tag{8}$$

holds under a single assumption: the calibration points and the test point are *exchangeable* [5, 48]. A random split satisfies this; a leave-group-out /

rolling-origin split does not—the calibration fold is drawn from the training groups and the test point from disjoint held-out groups, so the two obey different group-conditional laws and the nonconformity scores are no longer exchangeable across that boundary. The finite-sample guarantee is then void, and coverage can fall below $1-\alpha$ by the amount the score distribution shifts between groups—the covariate-shift / beyond-exchangeability regime of Tibshirani et al. [47] and Barber et al. [49], whose recovery of coverage requires a known or estimated inter-group likelihood ratio that plain split conformal does not supply. The observed collapse (section 3) is therefore the predicted consequence of applying an exchangeability-dependent procedure across a group boundary that breaks exchangeability; the uncertainty-ranked abstention repair makes no exchangeability claim and restores low conditional risk on the retained set without it.

Restoration bar and preregistered H1–H2.. A remedy r is said to restore coverage on a task t at split-repeat k only under the conjunctive bar used throughout the remedy audit: $\mathrm{cov}_{\mathrm{worst}} \geq 0.87$, a bootstrap CI on the reduction in Δ_{grp} versus split conformal that excludes zero, and no set-size blow-up. Write $\mathrm{Restore}(r,t,k) = 1$ if that bar is met. Let $\mathcal{T}$ be the evaluable held-out TableShift tasks under the frozen endpoint rule of section 3, K=20 the confirmatory split-repeats, and $m_{\mathcal{T}} = \lfloor |\mathcal{T}|/2 \rfloor + 1$ the majority threshold. The two hypotheses frozen before any confirmatory number were

$$\mathrm{H1}: \quad \min_{k \leq K} \left| \left\{ t \in \mathcal{T} : \exists r,\ \mathrm{Restore}(r,t,k) = 1 \right\} \right| \geq m_{\mathcal{T}} \tag{9}$$

(any tested remedy restores group-conditional coverage on a majority of evaluable tasks, at the minimum across K; this is the operational tally used in section 3), and

$$\text{H2}: \quad \min_{k\leq K}\left|\left\{t \in \mathcal{T} : \rho_u(t,k) \geq \rho_{\text{tuned}}(t,k)\right\}\right| \geq m_{\mathcal{T}} \tag{10}$$

(the free uncertainty score ties or beats a budget-matched 12-trial-tuned learned selector on a majority of evaluable tasks, robustly across K). H1/H2 are decided only on the held-out TableShift roster; the 14 keyed OpenML datasets that generated the hypotheses remain exploratory for this pair.

Reproducibility note.. All XGBoost/LightGBM numbers in section 3 are read from two reconciled runs—a Stage-1 XGBoost-only audit and a Stage-2 combined two-model audit—regenerated under a per-context deterministic RNG discipline (an independent random-number stream seeded per dataset $\times$ purpose $\times$ model for splits, calibration carve-outs, and bootstraps; see section 5 for the divergence this fixed). Under this discipline the two runs' XGBoost arms produce identical per-dataset statistics (verified on all 14 datasets), both GBM arms share identical splits and calibration folds, and reruns are deterministic. Result records carry provenance stamps (code version, seeds, package versions). Per-dataset one-sided bootstrap p-values are corrected across the 14 datasets with Holm–Bonferroni (FWER) and Benjamini–Hochberg (FDR) at $\alpha = 0.05$. Here m is the number of tests in the family (the 14 datasets), not the test-fold size n_{te} of (3): ordering $p_{(1)} \leq \cdots \leq p_{(m)}$, Holm rejects while

$$p_{(i)} \leq \frac{\alpha}{m - i + 1}, \tag{11}$$

and BH uses the step-up threshold $i\alpha/m$.

3. Results

We flag each result as *confirmatory* (it meets a preregistered per-arm bar) or exploratory/ descriptive (a statistic we report without asserting it as a robust claim). The abstention repair meets its preregistered per-arm bar robustly throughout, including on the held-out TableShift replication below (H2 CONFIRM), and is confirmatory; the grouped-gap count—the only preregistered bar on the degradation side—is underpowered and is accordingly reported as a descriptive distribution rather than a confirmatory count. Section 5 gives the dedicated rigor check behind that labeling choice: our conjunctive Stage-2 gate on the gap count evaluated to KILL under the reconciled protocol, so the confirmatory/exploratory labels used throughout this section are a post-hoc reinterpretation, not a claim of overall preregistered success. Our contribution on the degradation axis is therefore the split-realization robustness analysis itself, not the gate verdict; we report the count below as a distribution over split draws and read the collapse direction as a descriptive observation, not a separately preregistered bar.

Grouped splits collapse reliability (descriptive); abstention substantially improves it (repair: per-arm confirmatory).. Moving random→grouped degrades AURC suite-wide (e.g. Electricity AURC 0.031 → 0.224; base error 0.117 → 0.366), while split-conformal under-coverage is concentrated on temporal splits (4/6 temporal model×dataset pairs undercover—Electricity, Airline delays—vs. 4/22 grouped pairs), so the suite-wide collapse signal is the AURC one. Uncertainty-ranked abstention beats random deferral on 13 of 14 datasets (fig. 3); grouped splits raise AURC on the same suite (fig. 2). Descriptively, it cuts integrated selective risk by a median of 33% (XGBoost)

/ 34% (LightGBM) relative to random deferral (the repair ratio ρ of (6))—a substantial improvement, though (lacking persisted risk–coverage curves at matched coverage) we do not quantify what fraction of the random$\to$grouped gap this closes.

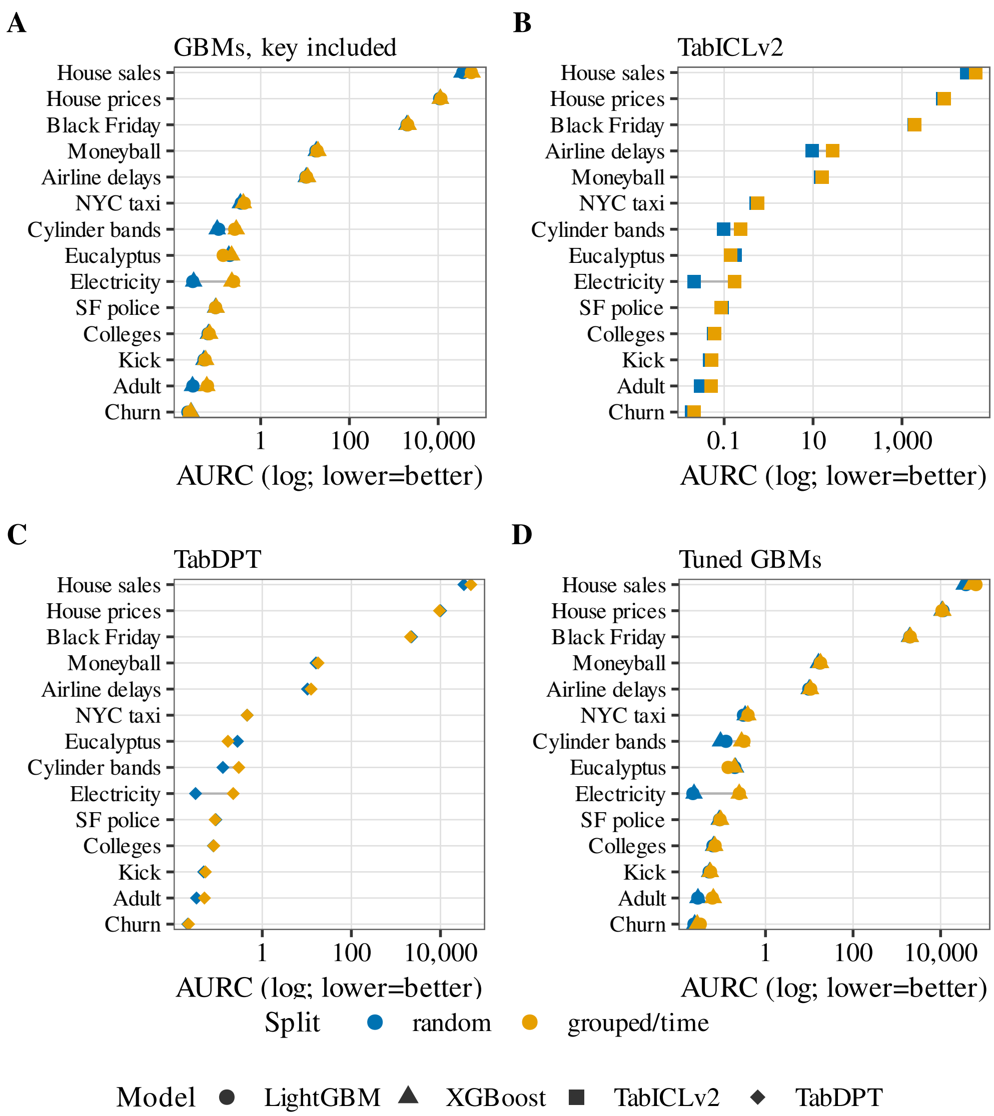

Figure 2: Per-dataset risk–coverage AURC under random vs. grouped/time split, shown as paired points on a log-scaled AURC axis (classification and regression AURC span several orders of magnitude). **A** GBMs with the group key included; **B** TabICLv2; **C** TabDPT; **D** tuned GBM baseline. Grouped/time points sit to the right of random on most datasets (higher AURC, worse selective risk).

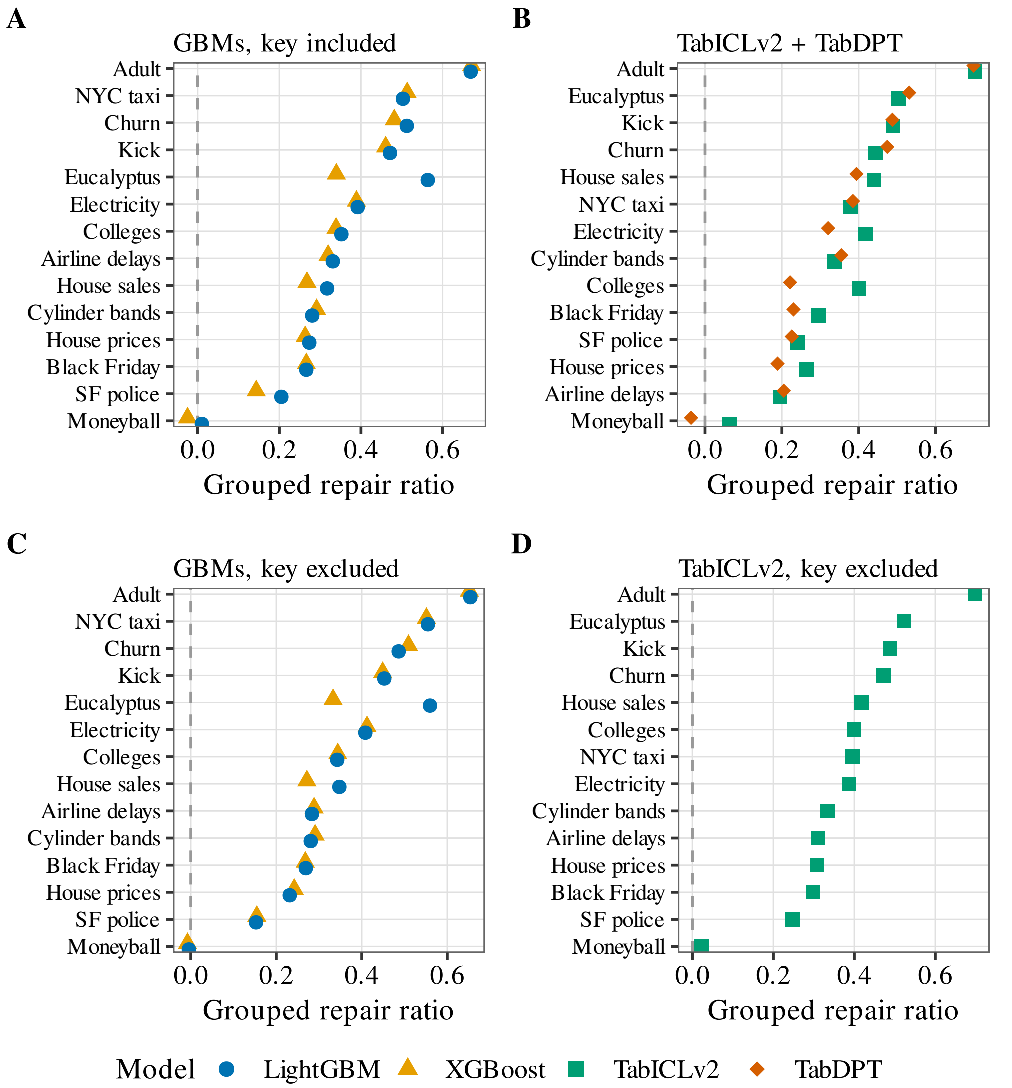

Figure 3: Uncertainty-ranked abstention beats random deferral under the grouped split. Each point is one dataset's repair ratio—the share of the random-deferral selective-risk AURC that abstention removes—with datasets ordered by ratio; points right of the dashed zero line improve on random deferral. **A** GBMs, key included; **B** TabICLv2 and TabDPT; **C** GBMs, key excluded; **D** TabICLv2, key excluded.

Model-agnostic across GBMs, with the repair robust and the count fragile.. On a single reconciled split the grouped reliability gap excludes zero on 8/14 (XGBoost) and 7/14 (LightGBM) datasets with the group key retained as a feature, or on the more conservative, less-confounded key-excluded design—which we argue in section 5 is the less inflation-prone of the two—on only 6/14 and 5/14, both below the preregistered majority threshold of 8; repair-beats-random holds 13/14 for both GBMs in both key conditions (fig. 3). We report both key conditions side by side here rather than headlining the higher (key-included) count, and note that under the less-confounded design the gap-count component is a minority result—which only reinforces that the gap count is not the primary claim. But that single-draw count is an *exploratory/descriptive* statistic, not the finding: over K=20–30 independent grouped split draws the grouped-gap count is a distribution with median 9/14 and 95% percentile interval $[6, 12]$ (XGBoost, key-included; LightGBM median 9, interval $[5, 11]$),[1] whose lower tail sits below the majority threshold—the gap count is majority-robust in only 75–83% of these split re-draws (which are independent re-partitions of the fixed 14-dataset panel, so the CI quantifies split-realization variance, not sampling over a dataset population; the 83% upper end is the Stage-1 XGBoost arm, the 75% lower end the Stage-2 LightGBM arm), so "a majority of datasets show a gap" is not a realization-robust claim (Fig. 5). The repair count, by contrast, is robust: its median is 13/14 and it exceeds the majority threshold in 100% of draws in every arm, with a 95% interval of $[12, 14]$ for the Stage-1 XGBoost arm (K=30, the

[1] Percentile-CI bounds are reported rounded to integer dataset counts (e.g. the XGBoost lower bound 6.7 is shown as 6); exact fractional bounds accompany the archived results.

same arm as the 83% gap figure above) and $[13, 14]/[12, 14]$ for the Stage-2 XGBoost/LightGBM arms (K=20). Hyperparameter tuning moves the single-draw count without a consistent sign (XGBoost 8 $\rightarrow$ 6/14, LightGBM 7 $\rightarrow$ 8/14; a wash on the majority question) while leaving repair at 13/14 (XGBoost) and 12/14 (LightGBM)—so the gap count is neither realization- nor hyperparameter-robust, whereas the repair is both. Multiple-comparison correction across the 14 datasets leaves the repair claim intact—13/14 datasets survive Holm–Bonferroni (11) for both GBMs—while the AURC-degradation tests survive Holm on 6/14 (XGBoost) and 5/14 (LightGBM) and BH-FDR on 7/14 and 6/14. (A key-excluded robustness ablation—dropping the leave-group-out/rolling-origin key from the model features—lowers the single-draw gap counts to 6/14 XGBoost and 5/14 LightGBM while repair-beats-random stays at 13/14 for both; see section 5.)

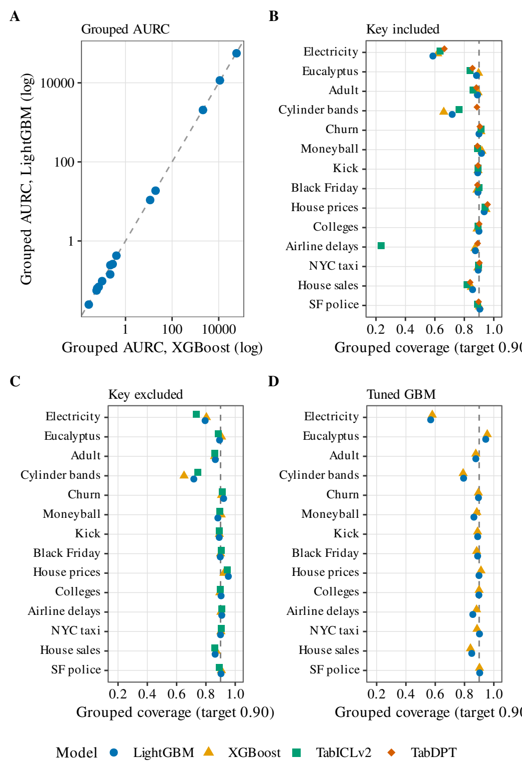

Figure 4: Model-agnosticism across four configurations spanning three architecture families. **A.** Grouped AURC for XGBoost versus LightGBM on all 14 datasets (log–log; points near the diagonal show the two gradient-boosted models behave near-identically). **B.** Split-conformal grouped coverage with the group key included, all four configurations; dashed line at target 0.90. **C.** The same coverage comparison with the group key excluded (XGBoost, LightGBM, and TabICLv2). TabDPT is omitted: this panel has no key-excluded coverage points for that configuration. **D.** Leakage-safe hyperparameter-tuned XGBoost and LightGBM; dashed line at target 0.90. Tuning moves the single-draw gap count without a consistent sign (XGBoost 8 $\to$ 6/14, LightGBM 7 $\to$ 8/14) and does not restore grouped coverage, while repair stays at 13/14 and 12/14. Both foundation models show the same collapse as the gradient-boosted models, with no auto-repair.

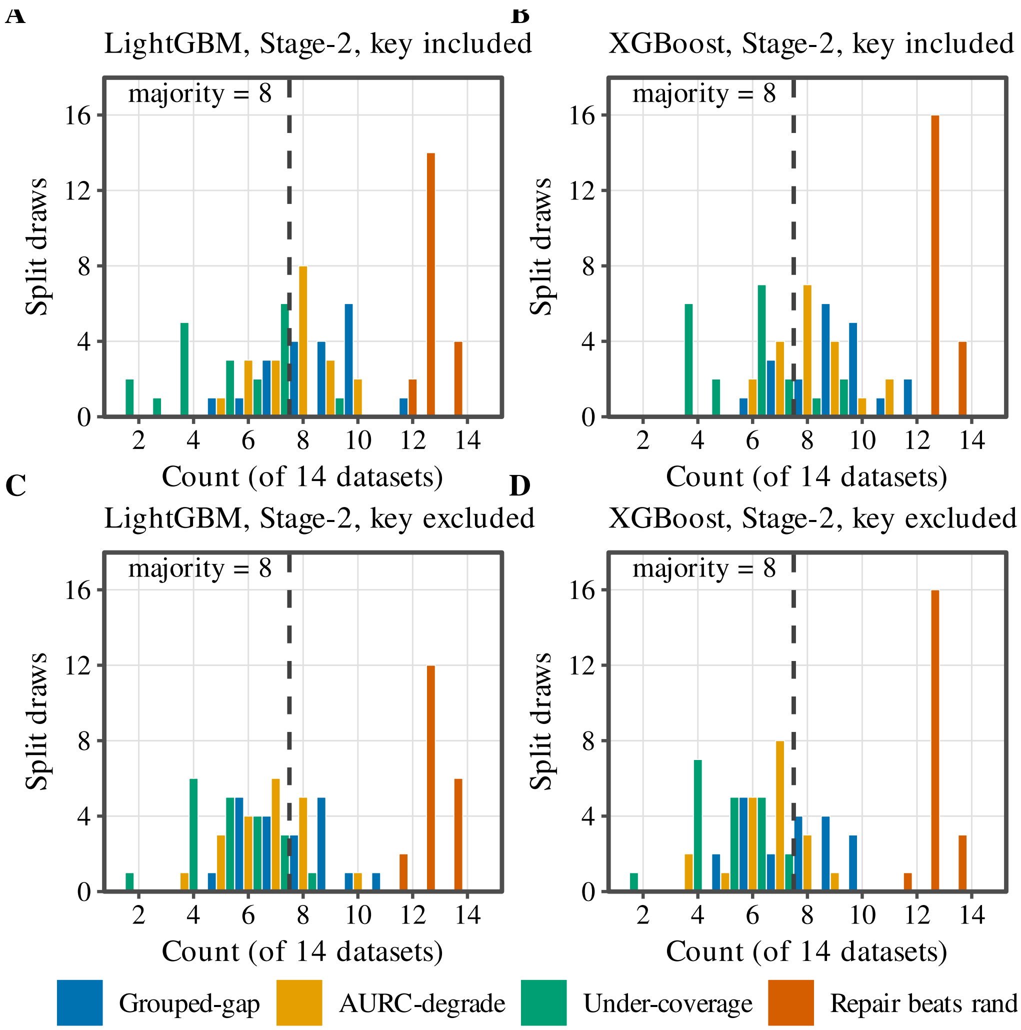

Figure 5: Split-realization fragility of the four headline counts over K repeated grouped-split draws (count out of 14 datasets; dashed line at the preregistered majority threshold of 8). **A.** LightGBM, Stage-2, key included. **B.** XGBoost, Stage-2, key included. **C.** LightGBM, Stage-2, key excluded. **D.** XGBoost, Stage-2, key excluded. The grouped-gap count straddles the threshold—majority-robust in 75% of LightGBM key-included draws (Stage-2 XGBoost 80%; combined range 75–83%)—AURC-degrade likewise straddles it, and under-coverage falls below it; only repair-beats-random sits at the top and never drops below the threshold (100% of draws). This is why we treat the repair, not the degradation-side counts, as the load-bearing finding.

Uncertainty-taxonomy Mondrian does not help (negative control).. The uncertainty-taxonomy Mondrian of (4) add-on improves grouped coverage on 12/28 model$\times$dataset pairs by point estimate but only 7/28 with CI excluding zero (median pooled improvement -0.002), well below the preregistered majority of 15 pairs, so it fails the Stage-2 Mondrian criterion. Its candidate mechanism, per-bin calibration fold size, is itself a case of the paper's split-sensitivity lesson: on a single frozen split under our reconciled per-context RNG discipline, the clean-threshold criterion (balanced accuracy ≥ 0.75, Mann–Whitney $p < 0.05$) passes in the key-included condition (bacc $= 0.854$, $p = 0.0011$, threshold 352 calibration observations; median per-bin size $\rho = 0.497$ [CI $0.10, 0.83$], minimum per-bin size $\rho = 0.518$, n_{cal} $\rho = 0.461$, all Spearman CIs excluding zero), but does not survive split-repeat robustness (median bacc $= 0.64$ [CI $0.55, 0.83$] across 20 re-draws, significant on only 4/20, median $p = 0.21$—the frozen 0.854/0.0011 is the favourable tail, not the typical case) and collapses entirely without the group/time key as a feature (zero predictors with CIs excluding zero, best separation bacc $= 0.548$, $p = 0.867$, chance level, preregistered verdict KILL; Fig. 6). We report the mechanism as real but split-fragile and key-dependent, not a confirmed explanation, per the paper's own split-sensitivity standard.

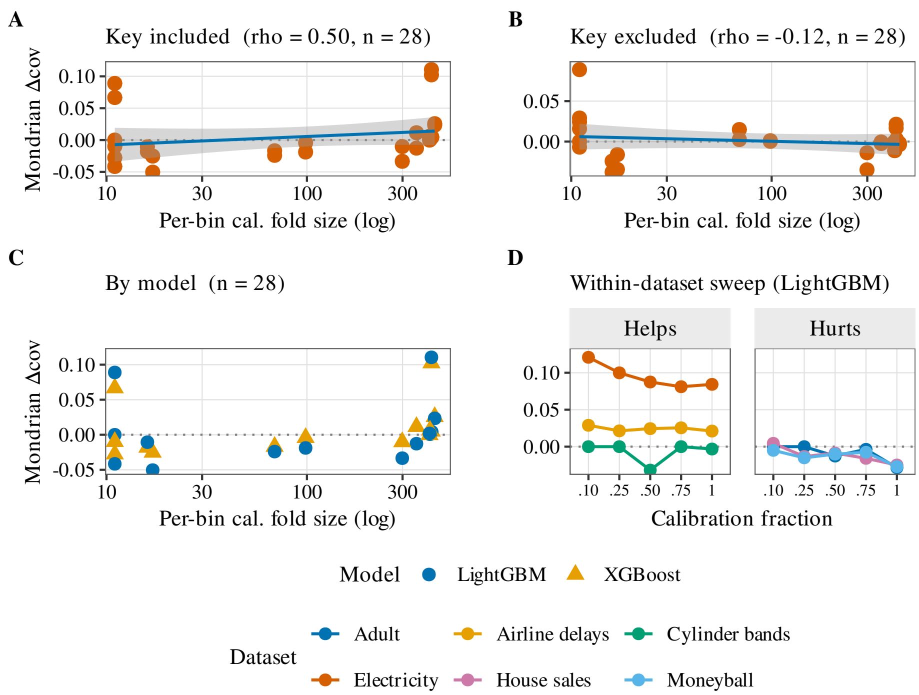

Figure 6: Why Mondrian conformal fails, and why that explanation is key-dependent (reconciled random-number stream). **A.** With the group key included, per-bin calibration-fold size predicts Mondrian coverage improvement (Spearman $\rho = 0.50$, CI excluding zero; clean-threshold separation bacc $= 0.854$, $p = 0.0011$, $n = 28$)—the mechanism passes the preregistered bar. **B.** Dropping the group key from the model features collapses the relationship ($\rho = -0.12$, no significant predictors, threshold bacc $= 0.548$, $n = 28$), flipping the preregistered verdict to KILL. **C.** The same $n = 28$ pairs as A, coloured by model (LightGBM, XGBoost): the key-included association is not confined to one booster. **D.** Within-dataset calibration-fraction sweep (LightGBM), split into datasets where Mondrian helps versus hurts: raising the calibration fraction does not uniformly improve coverage, and can worsen it where Mondrian already hurts. The calibration-size mechanism is real but key-dependent.

Is the repair a low bar? A learned selector does not beat the free uncertainty score

To check that abstention's advantage over random deferral is not merely the low bar of beating a coin flip, we compare the free model-native uncertainty score against a learned post-hoc selector—a small untuned (default-hyperparameter, no search) gradient-boosted regressor trained on the calibration fold to predict per-example loss from prediction-level features (max-probability, entropy, margin for classification; interval width and endpoints for regression), using its predicted risk as the abstention score—in the identical risk–coverage frame (same test folds, calibration carve-out, and AURC functions; the selector never sees the test fold). Table 1 summarizes the head-to-head over the 14 keyed datasets under the grouped split (XGBoost). The free uncertainty score is not a low bar: it reproduces the repair (beats random deferral on 13/14) and matches the trained selector, with statistically indistinguishable median repair ratios, both far below the oracle ceiling (Table 1; oracle ranks by the true loss). Throughout, the *repair ratio* is ρ in (6) (higher is better; zero means no improvement over random deferral). In the paired head-to-head the uncertainty score wins on 8/14 datasets and is never beaten on classification; the learned selector wins on only 3/14, all regression (Airline delays, NYC taxi, House sales), where it can exploit interval geometry beyond raw width, with 3 ties. The learned selector also needs calibration errors to fit, which dominant-group splits barely supply (Adult and Cylinder bands leave only 56 and 45 calibration rows), a structural disadvantage the free score does not share. We scope this claim: the selector is untuned and gets none of the 12-trial hyperparameter budget our GBMs receive, so the head-to-head is asymmetric in the free score's favour, and the pattern is better read as "the free score wins where calibration errors are scarce (classification, small folds) and the selector wins where they are abundant (the three data-rich regression datasets)" than as a global tie—the equal medians average over those two regimes. We tested the obvious objection directly: giving the learned selector the same 12-trial randomized-search budget the GBMs receive (a budget-matched, exploratory head-to-head over $K{=}10$ grouped split-repeats with $N{=}400$ paired bootstraps) does not overturn the verdict—the free uncertainty score still ties-or-beats the budget-matched tuned selector on an aggregate median of 11.5 of the 14 datasets (min 9, max 13 across the ten split-repeats), and at the anchored single draw (k_0) on 9 of the 12 non-degenerate datasets (the remaining 2—Adult and Cylinder bands—excluded because the tuned selector is untrainable on their dominant-group calibration fold). Our claim is therefore the weaker, defensible one—the free uncertainty score is not a low bar (it is never beaten on classification, ties on median, and is not overturned by a budget-matched tuned selector), not that no learned selector can ever win. This arm is XGBoost-only; LightGBM behaves near-identically throughout, so the verdict transfers.

Foundation-model arms: TabICLv2 and TabDPT collapse the same way

To test whether a tabular foundation model auto-corrects the reliability gap, we ran two open-weights, token-free tabular FMs—TabICLv2 [18] (ensemble of 16) and TabDPT [19] (8 ensemble members)—under the identical split-conformal + abstention protocol on all 14 keyed OpenML datasets, each with a per-context deterministic RNG and the reconciled Stage-1 GBM reference. Table 2 reports the TabICLv2 (leakage-ablated) numbers; Fig. 4B

Table 1: A learned post-hoc abstention selector vs. the free uncertainty score vs. random deferral (grouped split, 14 keyed datasets, XGBoost). Head-to-head is a paired test-set bootstrap on the repair-ratio difference; "wins" means the difference CI excludes zero. Numbers from the learned-selector comparison.

Comparison / statistic	value
uncertainty abstention beats random deferral	13/14
learned selector beats random deferral	12/14
head-to-head: uncertainty wins	8/14
head-to-head: learned selector wins	3/14 (all regression)
head-to-head: tie	3/14
median repair ratio — uncertainty	0.339
median repair ratio — learned selector	0.338
median repair ratio — oracle (true-loss ranking)	0.745

shows the four-model coverage comparison.

Table 2: TabICLv2 FM arm, leakage-ablated (ensemble of 16, key retained), all 14 datasets: split-conformal coverage ($\alpha = 0.10$, target 0.90) and grouped repair ratio under random vs. grouped/time split. Numbers from the leakage-ablated TabICLv2 arm; 400-bootstrap CIs. Undercover= grouped coverage < 0.88 (target -0.02 slack). The gap column is the coverage-gap point estimate (cov R $-$ cov G computed before rounding), so it can differ from the displayed three-decimal subtraction by one unit in the last place.

Dataset	key	kind	cov R	cov G	gap	repair ratio (grouped)
Electricity	date	time	0.898	0.634	+0.264	0.417 [0.396, 0.439]
Eucalyptus	Year	time	0.904	0.838	+0.067	0.503 [0.400, 0.588]
Adult	native-country	group	0.902	0.858	+0.043	0.703 [0.696, 0.710]
Cylinder bands	paper mill location	group	0.876	0.765	+0.111	0.337 [0.219, 0.451]
Churn	state	group	0.907	0.913	−0.006	0.443 [0.247, 0.593]
Moneyball	Team	group	0.905	0.889	+0.016	0.064 [−0.011, 0.130]
Kick	ZIP	group	0.895	0.893	+0.002	0.489 [0.447, 0.525]
Black Friday	city category	group	0.905	0.899	+0.007	0.295 [0.287, 0.303]
House prices	Neighborhood	group	0.952	0.940	+0.013	0.264 [0.204, 0.313]
Colleges	state	group	0.914	0.891	+0.023	0.400 [0.381, 0.421]
Airline delays	DayofMonth	time	0.906	0.234	+0.672	0.195 [0.184, 0.204]
NYC taxi	pickup location	group	0.901	0.895	+0.006	0.378 [0.358, 0.400]
House sales	zipcode	group	0.909	0.817	+0.092	0.440 [0.422, 0.458]
SF police	district	group	0.894	0.891	+0.003	0.241 [0.194, 0.285]

The FM collapses under group/time shift across the full 14-dataset suite: for TabICLv2, AURC degrades (CI excluding 0) on 6/14 datasets, grouped conformal coverage falls below the literal 0.90 target on 12/14 datasets (median grouped coverage 0.890; on the stricter count that requires missing the target by more than the 0.02 slack used in Table 2, 6/14), and abstention-repair beats random deferral on 13/14 (median grouped repair ratio 0.389). Dropping the group key from the FM's features (leakage ablation) leaves the repair at 13/14 and lowers the AURC-degrade count to 4/14, consistent with the GBM key ablation.

A second FM replicates it.. TabDPT shows the same pattern: repair beats random on 13/14 datasets (both key conditions), grouped AURC degrades on 5/14, and median grouped coverage is 0.890—so the collapse-plus-repair story now reproduces across four configurations spanning three architecture families (GBDT plus two distinct tabular-FM architectures; Fig. 4B), no FM auto-repairing the gap. The hardest shifts remain temporal: Airline delays grouped coverage collapses to 0.234 and Electricity to 0.634 (TabICLv2), comparable to or worse than the reconciled XGBoost baseline, confirming that the FMs require the same conformal + abstention recipe—not an auto-repair. These leakage-ablated ensemble-of-16 numbers, with the reconciled GBM reference, are the headline TabICLv2 results; a smaller-ensemble configuration agrees qualitatively on the shared classification datasets and is not used for any headline number (see section 5).

Robustness checks.. A 30-dataset breadth extension (the 14 keyed datasets plus 16 OpenML datasets grouped on an arbitrary categorical feature, so the "grouped shift" is less meaningful there) keeps abstention-repair ahead of random on 25/29 completed datasets for TabICLv2 (the single non-completing dataset is Phishing websites, excluded before evaluation because no valid grouping key could be assigned), and an in-context scaling sweep (ensemble $\times$ context length, 64 cells) shows AURC improving smoothly with more context without altering the grouped-collapse conclusion; we report both as appendix-level robustness, not as the keyed-suite headline.

Generalization to a TableShift-class benchmark (Folktables/ACS)

We test whether the model-agnostic finding survives on a canonical distribution-shift suite. TableShift's 11 open tasks are publicly downloadable. Two of them, Assistments and College scorecard, require Kaggle credentials and were excluded before any confirmatory number was computed, leaving 9 open tasks that enter the generalization test reported below, which resolves the held-out out-of-distribution test fold's often-single-domain endpoint ambiguity with a frozen evaluability rule fixed ahead of time. For the present, complementary generalization arm we use the open TableShift-class substitute [2]: four Folktables/ACS [50] binary tasks (ACS Income, ACS Public Coverage, ACS Mobility, ACS Employment) over US Census ACS PUMS 2018, with a leave-states-out spatial OOD split (held-out {MA, GA, MI}) against a random in-domain split (XGBoost + LightGBM, split-conformal $\alpha = 0.10$, 400-bootstrap CIs).

Abstention repair replicates.. Uncertainty-ranked abstention beats random deferral under OOD on 4/4 tasks for both GBMs, every CI excluding 0 (median OOD repair ratio 0.539 XGBoost, 0.579 LightGBM; Fig. 7A).

The OOD reliability gap manifests as a subgroup coverage gap (exploratory generalization).. We flag this arm as exploratory: the primary, preregistered marginal criterion (AURC-degrade or marginal under-coverage) replicated on only 1/4 tasks per GBM under this milder spatial shift, so the finding below rests on a post-hoc subgroup criterion and should be read as hypothesis-generating, not confirmatory. On this milder ACS spatial shift the marginal-AURC degradation is weak (only 1/4 tasks per GBM degrade with CI> 0; median AURC Δ 0.003 XGBoost, -0.002 LightGBM); instead the gap appears as a RAC1P race-subgroup conformal coverage gap—worst-minus-best is negative on all 8 (model, task) cells, with the gap CI excluding 0 on 4/4 tasks for both GBMs (median -0.033 XGBoost, -0.027 LightGBM; Fig. 7B)—but this 4/4 is the post-hoc, broadened subgroup criterion, not the preregistered marginal one. Only the repair generalizes to this milder ACS shift (the abstention improvement replicates 4/4); the marginal reliability collapse does not (AURC-degrade / under-coverage on just 1/4 tasks per GBM), and the RAC1P subgroup-coverage-gap 4/4 survives only because the success criterion was broadened after the marginal one missed—so the collapse itself is benchmark-dependent and does not generalize here, only the repair does.

Note on the reliability-gap criterion.. The OOD reliability-gap criterion here is a 3-way disjunction (AURC-degrade CI> 0 OR marginal under-coverage OR RAC1P subgroup-coverage-gap CI< 0), broadened from the primary 14-dataset study's 2-way disjunction (AURC-degrade OR coverage-gap only), because the narrow criterion alone replicated on only 1/4 tasks per GBM under this milder spatial shift.

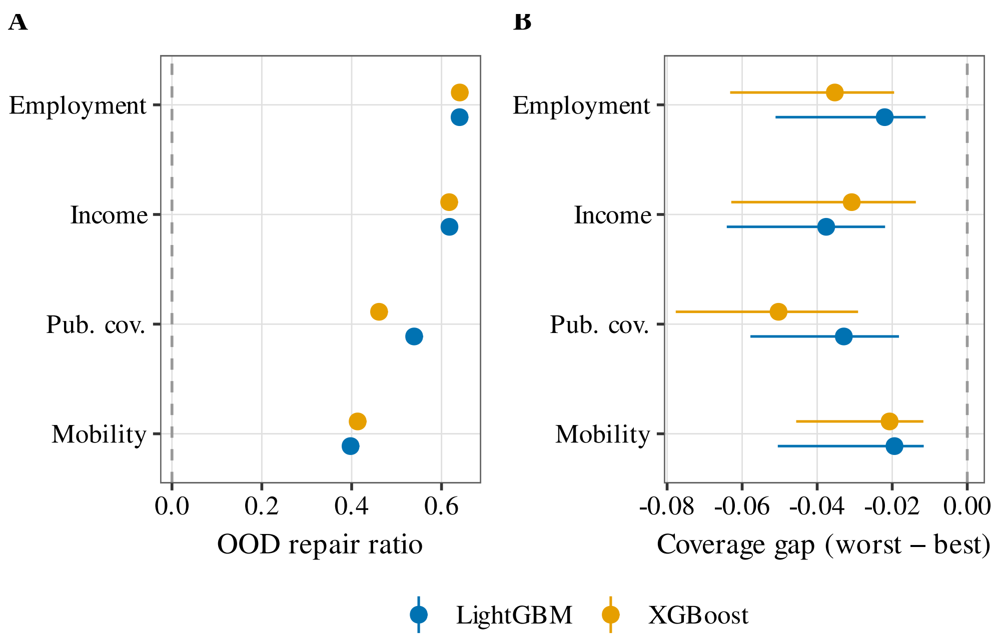

Figure 7: Generalization to a TableShift-class benchmark (Folktables/ACS, leave-states-out spatial OOD). **A.** OOD abstention repair ratio per ACS task (XGBoost vs. LightGBM, 400-bootstrap CIs); the repair ratio is the fractional reduction in integrated selective risk vs. random deferral, so positive ratios (CI excluding 0) mean uncertainty abstention beats random deferral, replicating on 4/4 tasks for both GBMs. **B.** RAC1P race-subgroup conformal coverage gap (worst−best); the OOD reliability gap manifests as a negative subgroup coverage gap (CI excluding 0 on 4/4 for both GBMs) rather than a marginal-AURC blow-up.

Exploratory audit: do distribution-free remedies restore group-conditional coverage? (No.)

We close the diagnostic by asking, exploratorily, whether the coverage side of the collapse can be repaired by a *distribution-free* conformal remedy that never sees the (absent) grouping key. We label this arm exploratory throughout, per its own preregistration: it re-uses the 14 keyed OpenML datasets that generated the hypothesis, so the confirmatory test is reserved for a held-out TableShift family the exploratory data cannot contaminate, and nothing here is read as confirmatory. It is run at K=10 grouped split-repeats—the preregistration specifies K=20 for the generalization version; we report K=10 here as the exploratory budget and flag the difference—with N=400 paired-bootstrap CIs and the GBM loss and uncertainty arrays identical to the frozen backbone.

The disease.. Under the grouped / rolling-origin split, plain split conformal—the paper's headline recommendation—under-covers group-conditionally: median worst-group coverage 0.810 against the 0.90 target, median worst−best group gap 0.144. The marginal guarantee holds; the group-conditional one does not—the exchangeability break made concrete.

The remedy evaluation.. We run the two standard distribution-free remedies—weighted / covariate-shift CP [47] and inferred-group Mondrian (group-conditional CP on KMeans clusters of the covariate space, the split key dropped)—plus the preregistered KILL null (the uncertainty-decile Mondrian). A remedy restores coverage on a dataset only under a conjunctive bar: worst-group coverage $\geq$ 0.87 and a bootstrap-CI-positive group-gap reduction vs. split conformal and no set-size blow-up. The count of the 14 datasets where any remedy clears that bar has median 1 (min 0, max 3) across the ten split-repeats—never a majority (threshold 8). Per remedy, the median worst-group coverage at the anchored draw (k_0) is 0.831 (weighted), 0.817 (inferred-Mondrian) and 0.811 (unc-decile KILL null)—all below the 0.87 restoration threshold, so the negative already holds on the worst-group point estimate alone, before the conjunctive CI and set-size conditions are even applied.

The mechanism.. This is not a failure of the remedies in general: weighted CP provably restores coverage under pure covariate shift, and inferred-Mondrian restores worst-group coverage under subpopulation heterogeneity—both are validated on synthetic data constructed so those assumptions hold exactly. Their failure here indicates that leave-group-out induces a shift beyond covariate-only (a label/concept component the covariate-space reweighting cannot capture), or from calibration-fold scarcity our design cannot separate, so we scope the negative narrowly to the two standard distribution-free remedies we implemented. The stronger subpopulation-shift and audited-CP estimators [29, 30] are not run here; we cite them as the open coverage-restoring question, not as implied-refuted. The prescription is therefore negative for coverage, positive for selection: no tested distribution-free remedy restores group-conditional coverage without the group key, while the free uncertainty score reliably ties-or-beats a budget-matched tuned selector (above).

Confirmatory coverage-remedy evaluation on the held-out TableShift roster

We now report the held-out evaluation for which the exploratory remedy audit above reserved its hypotheses: a preregistered test of H1 and H2 on TableShift's own open task roster, disjoint from the 14 keyed OpenML datasets that generated the two hypotheses. Frozen ahead of any confirmatory number, and written as (9)–(10): **H1 (coverage restoration)**—some single remedy restores group-conditional coverage on a majority of evaluable held-out tasks, at the minimum across K=20 split-repeats; **H2 (selective prescription)**—the free uncertainty score ties or beats a budget-matched, 12-trial-tuned learned selector on a majority of evaluable tasks, robustly across K.

Roster and frozen endpoint rule.. All 9 open (credential-free) tasks complete (9/9; K=20 split-repeats each, N=400-bootstrap CIs). A single rule, frozen and applied mechanically before any confirmatory number was computed, assigns each task exactly one endpoint (Table 3): 3 tasks have too few domain groups for any group-conditional endpoint and are excluded-by-rule from the H1/H2 tally; the remaining 6 evaluable tasks split into 2 with a primary worst-group-domain endpoint and 4 with a fallback marginal-gap-plus-secondary-grouping endpoint. That split was chosen by the same frozen rule, and the reader should weigh it accordingly: 4 of the 6 evaluable tasks certify under the fallback (marginal-gap plus secondary-grouping) endpoint rather than the primary worst-group-domain endpoint that motivates the study, so the roster-level H1/H2 verdicts below are majority-carried by the fallback construct, not the primary one. The rule is endpoint-blind—it selects an endpoint from each task's own group structure, not from which answer that endpoint would produce—so the fallback-heavy composition is a property of the roster, not a chosen-to-flatter selection; in short, the generalization yield below is one confirmed negative on coverage (H1 REJECT) plus one confirmed tie on selection (H2 CONFIRM), not a positive method win.

H1: no remedy robustly restores group-conditional coverage (REJECT).. Under the leave-group-out split, plain split conformal under-covers group-conditionally on the evaluable roster (median worst-group coverage 0.797 against the 0.90 target, median worst−best group gap 0.135), replicating the

Table 3: Frozen per-task endpoint decisions on the 9-task open, held-out TableShift roster (evaluability rule applied mechanically, before any confirmatory number was computed). The 3 excluded-by-rule tasks report only a descriptive marginal in-domain versus out-of-domain coverage gap; the 6 evaluable tasks enter the H1/H2 confirmatory tally.

Task	Endpoint	n groups	In H1/H2 tally
Diabetes readmission	marginal gap only	0	excluded-by-rule
NHANES lead	marginal gap only	0	excluded-by-rule
PhysioNet	marginal gap only	0	excluded-by-rule
BRFSS diabetes	primary worst-group domain	5	yes
BRFSS blood pressure	fallback (secondary: age group)	7	yes
ACS Income	fallback (secondary: age)	71	yes
ACS Public coverage	fallback (secondary: age)	50	yes
ACS Food stamps	fallback (secondary: age)	44	yes
ACS Unemployment	primary worst-group domain	15	yes

exploratory audit's diagnosis on a fully held-out benchmark. The count of evaluable tasks where any remedy clears the restores-coverage bar has minimum 3, median 3, maximum 4 across the 20 split-repeats—never reaching the majority threshold of 4 at the minimum, so H1 is REJECTed. Applying the same fragility scrutiny used elsewhere in this paper to the 14-dataset gap count, this REJECT itself rests on only 6 evaluable tasks (4 of 6 on the fallback endpoint, Table 3) and is not robust to which task is held out: recomputing the frozen $\lfloor n/2 \rfloor + 1$ majority rule at $n{=}5$ (leave-one-task-out) flips the verdict to CONFIRM if any of the three tasks where no remedy ever restores coverage across K (BRFSS diabetes, BRFSS blood pressure, ACS Unemployment) is dropped, and leaves it REJECT if any of the three tasks where a remedy restores coverage at every repeat (ACS Income, ACS Public Coverage, ACS Food Stamps) is dropped. Per remedy (min/median/max across K): weighted CP 3/3.0/4, inferred-group Mondrian 0/0.0/0, uncertainty-decile Mondrian 0/0.0/1 (Table 4)—the same ordering as the exploratory audit, with weighted CP strongest and still short of a robust majority. A supplementary, non-decision-altering multiplicity correction (Holm–Bonferroni / Benjamini–Hochberg across the 6 evaluable tasks, anchored single-draw p-values) rejects the gap-reduction null on 5/6 tasks at $\alpha = 0.05$ (all but ACS Unemployment, $p = 0.165$): individual remedies do measurably shrink the gap on most tasks, just not far enough, and not consistently enough across K, to clear the conjunctive restoration bar on a majority.

H2: the free uncertainty score ties-or-beats the tuned selector on every task (CONFIRM).. The uncertainty-ranked selector ties or beats the 12-trial-tuned learned selector on all 6/6 evaluable tasks, at every one of the $K{=}20$ split-repeats (minimum = median = maximum = 6)—a fully robust replication of the exploratory audit's finding, with no exception across any task or repeat. The matching supplementary multiplicity correction on the tuned-minus-uncertainty repair-ratio delta finds $p = 1.000$ on all 6 tasks (the tuned selector never wins). H2 is CONFIRMed.

Table 4: Confirmatory H1/H2 verdicts on the 6 evaluable, held-out TableShift tasks (K=20 split-repeats, N=400-bootstrap CIs). Min/median/max are computed across the 20 split-repeats; majority threshold= 4.

Comparison / statistic	value
median split-conformal worst-group coverage (target 0.90)	0.797
median split-conformal worst−best group gap	0.135
any-remedy restores coverage (min / median / max across K)	3 / 3.0 / 4
weighted CP	3 / 3.0 / 4
inferred-group Mondrian	0 / 0.0 / 0
uncertainty-decile Mondrian	0 / 0.0 / 1
H1 (coverage restoration)	**REJECT**
uncertainty ties-or-beats tuned selector (min / median / max across K)	6 / 6.0 / 6
H2 (selective prescription)	**CONFIRM**

Descriptive: the marginal in-domain versus out-of-domain coverage gap where available.. TableShift exposes an additional in-domain held-out test split for only 3 of the 9 tasks; where available, the marginal (non-group-conditional) coverage gap under plain split conformal is PhysioNet +0.094, BRFSS diabetes +0.040, and Diabetes readmission −0.018—two of three tasks show the same direction (out-of-domain coverage below in-domain) as the primary 14-dataset study's headline collapse, with Diabetes readmission's small negative gap the exception.

Disclosed loader adaptations.. Applying the frozen endpoint rule and remedy ladder to TableShift required three narrow adaptations that do not change the frozen decision rule or the remedies: (i) passing the ACS domain column through TableShift's standard (otherwise empty) preprocessing; (ii) correcting a case-sensitivity mismatch in the NHANES download so the intended survey wave is used; (iii) reading only the required columns from the wide ACS person and household files. Each adaptation was checked by reproducing an already-completed task with identical numerical results before being applied to the remaining tasks; the frozen evaluability and endpoint rule itself was never modified.

4. Discussion

The Introduction asked whether random-split validation overstates reliability under grouped and temporal deployment [1, 2, 3], and whether a free uncertainty score is a defensible selective-risk fallback when split-conformal coverage (8) fails. The answers, read against the same H1/H2 pair (9)–(10) and the same tables and figures, are as follows.

(i) *Directional collapse, fragile count.* Grouped splits degrade AURC (5) suite-wide (fig. 2) and push split-conformal coverage (3) below the 0.90 nominal target on temporal splits, including both tabular foundation models (fig. 4B). That is the predicted consequence of breaking exchangeability [5, 48, 47, 49], not a surprise relative to the guarantee in section 2. The dataset-count of the gap is not a preregistered majority claim: it straddles the threshold across split draws (fig. 5) and is key- and hyperparameter-fragile (section 5).

We therefore read the degradation as a direction, matching the Introduction, not as a confirmed majority finding.

(ii) *Abstention repairs selective risk, not coverage.* Uncertainty-ranked abstention beats random deferral on 13 of 14 datasets (fig. 3), with median repair ratio ρ (6) of 0.339 against an oracle ceiling of 0.745 (table 1). On the held-out TableShift roster [2] the free score ties-or-beats a budget-matched tuned selector on all 6 evaluable tasks at every one of $K=20$ repeats (table 4), so H2 is CONFIRMed. Abstention makes no coverage claim and does not restore (8); the Airline delays coverage collapse to 0.234 is not repaired by it. The Introduction's warning that reject options can shape subgroup disparities [12, 13] returns in the Folktables/ACS arm as a RAC1P race-subgroup coverage gap (fig. 7).

(iii) *The effect is model-agnostic, and neither FM auto-repairs it.* The two GBMs behave near-identically (fig. 4A). TabICLv2 [18] and TabDPT [19] show the same grouped/temporal collapse as XGBoost and LightGBM (fig. 4B), contrary to the hypothesis that a tabular foundation model would auto-correct shift [24, 22, 20, 21]. We did not run license-gated TabPFN v2 [23]; the token-free roster is the one the Introduction advertised.

(iv) *The uncertainty-taxonomy Mondrian does not help.* Per-bin quantiles (4) fail the Stage-2 majority bar, and the calibration-size explanation is split-fragile and key-dependent (fig. 6), exactly the negative control previewed in the Introduction.

H1 is REJECTed on the same held-out roster: no tested distribution-free remedy restores group-conditional coverage on a majority of evaluable tasks at the minimum across K (table 4; (9)). Weighted / covariate-shift conformal (7) is the strongest of the two standard remedies and still short of a robust majority; inferred-group Mondrian never restores. Stronger subpopulation- and audited-CP estimators [29, 30] remain untested here. Adaptive conformal inference [36] and the jackknife+ [38] likewise sit outside the audit. The recipe we recommend is therefore the one the Introduction scoped: plain split-conformal plus uncertainty abstention, for retained-set selective reliability [7, 8, 31, 32], not as a coverage fix.

What we deliberately do not claim is a stable "majority of datasets show a gap" count: over repeated split draws, hyperparameter tuning, and key inclusion, that count ranges over roughly 5–12 of 14 and is majority-robust in only three-quarters to four-fifths of draws with the group key included, falling to roughly half with the key excluded (section 5), whereas the repair beats random deferral on 13/14 in every arm and every draw. The uncertainty-taxonomy Mondrian add-on is not a free win: it needs large per-stratum calibration, which grouped-shift tabular data rarely provides. The calibration-size explanation for that failure (section 3, fig. 6) clears our preregistered bar only on a single frozen split with the group key retained, does not survive our own split-repeat robustness check, and collapses to chance once the key is dropped; we therefore treat it as a split-fragile, key-dependent association rather than an established mechanism, and recommend reporting all conditions rather than the favorable one alone.

5. Limitations

This section collects the rigor check: the preregistered Stage-2 gate's KILL verdict, the split-realization fragility behind Fig. 5, and the key-inclusion sensitivity of the headline counts are stated here once, in full, rather than scattered across the paper, because the scope of a preregistered study belongs in one place a reader can check against the confirmatory claims above.

Scope and future work.. The 14 keyed OpenML datasets are a convenience suite; we have already extended to the open TableShift-class benchmark (Folktables/ACS) above, where the abstention-repair finding replicates and the OOD reliability gap surfaces as a RAC1P subgroup coverage gap. As noted above (section 3), the ACS reliability-gap criterion was broadened to a 3-way disjunction because the narrow 2-way criterion alone replicated on only 1/4 tasks per GBM under this milder spatial shift. The FM roster covers all 14 keyed datasets for both TabICLv2 (leakage-ablated, ensemble of 16, reconciled GBM reference) and TabDPT. The FM roster still omits TabPFN-v2/2.5 [23, 51], which we leave for future work pending license review; we defend the current roster on license-cleanliness (all four configurations are open and token-free). The credentialed TableShift medical tasks (e.g. MIMIC/ICU) remain future work behind their data licenses.

Cross-run reproducibility of XGBoost — root-caused and fixed; the cost is one count revision.. The two independent XGBoost runs behind earlier drafts (the Stage-1 and Stage-2 audits) diverged sharply on some datasets (e.g. Adult grouped repair-ratio 0.635 vs. 0.119; grouped-gap count 8/14 vs. 10/14) despite fully seeded, internally deterministic execution. The cause was a shared random-number stream: both analyses drew every stochastic step from a single generator, so Stage-2's second (LightGBM) arm advanced the stream before the next dataset's splits were drawn. Dataset #1 (Electricity) matched exactly across the two runs while every later dataset diverged, confirming the two runs were evaluating different split realizations rather than exhibiting model irreproducibility. We replaced the shared stream with per-context deterministic RNGs (seeded per dataset $\times$ purpose $\times$ model): the two XGBoost arms now agree exactly on all 14 per-dataset records, and both GBMs share identical splits and calibration folds. Under the reconciled protocol the headline counts are those reported in section 3 (gap 8/14 XGBoost, 7/14 LightGBM; repair 13/14 both). LightGBM's gap count now sits one dataset below the preregistered majority threshold, so the conjunctive Stage-2 gate (grouped-gap majority and Mondrian majority) evaluates to KILL under the reconciled protocol even though the repair sub-criterion replicates strongly; the confirmatory/exploratory split used throughout is accordingly a post-hoc reinterpretation, not an overall preregistered success (section 3). The gate is underpowered on the gap arm: Stage-1 (XGBoost) passed it at exactly the majority threshold (8/14, a knife-edge), a single-dataset flip is enough to fail it, and single-draw counts across the pre- and post-fix realizations have in fact ranged 7–10 of 14—fig. 5 and section 3 give the full split-realization distribution behind that fragility. A verdict this sensitive to one re-draw carries little information, so we treat the Holm-robust repair as the confirmatory finding and the degradation direction as descriptive, not the gate outcome itself. The headline FM arm (ensemble of 16, leakage ablation) uses the reconciled Stage-1 GBM reference; a smaller-ensemble configuration that retained the pre-reconciliation reference is not cited for any headline number.

Group-key feature leakage — bounded by a key-excluded ablation.. In the main reported runs the leave-group-out/rolling-origin key (e.g. `native-country`, `date`) is retained as a label-encoded model feature. The Stage-2 preregistration described this choice as conservative for the degradation claim, on the reasoning that seeing the key would make grouped prediction easier; that reasoning understates the risk in the other direction. Because leave-group-out test keys take categorical values never seen during training, a model that has learned to rely on key-specific patterns in-sample can behave erratically out-of-sample on the encoded key alone, so retaining the key can inflate, not just dampen, the apparent grouped-split collapse—confounding genuine group/time shift with an ID-generalization artifact. We re-ran the Stage-1 and Stage-2 analyses with the split key dropped from the feature encoder (the key-excluded runs); we also re-ran both FM arms under the key-excluded ablation (TabICLv2 and TabDPT, repair 13/14 in both conditions), while the TableShift/ACS arm is out of scope for this specific Table 1 headline (its own separate reliability-gap criterion is discussed above). Under the reconciled RNG protocol, excluding the key lowers the grouped-gap count from 8/14 to 6/14 (XGBoost; identical in the Stage-1 and Stage-2 runs, which now agree exactly) and from 7/14 to 5/14 (LightGBM)—below the preregistered majority threshold of 8 for both GBMs—while the repair-beats-random count is unchanged at 13/14 for both. We read this as evidence that key leakage inflates the grouped-gap count (roughly two datasets' worth for each GBM), on top of the split-realization sensitivity documented above. The robust core claim is therefore the repair result—uncertainty abstention beats random deferral on 13/14 datasets in every arm (key-included, key-excluded, both GBMs, and both FMs; the GBM/FM correction asymmetry behind that shorthand is detailed below)—while the grouped-degradation majority count should be read as realization- and design-sensitive (range 5–8 of 14 across the reconciled key-included/-excluded runs) pending repeated-split-draw CIs. Section 3's headline numbers cite the key-included reconciled runs; the key-excluded counts above are the conservative lower bound.

FM repair claims rest on CIs, not the Holm-corrected bar (limitation).. The GBM repair claim clears Holm–Bonferroni across all 14 datasets; the two FM arms' repair claim does not undergo the same correction, because per-dataset p-values for a Holm pass on the FMs were not persisted, so the FM 13/14 is corroborated only by 400-bootstrap CIs excluding zero—a CI-level, not a FWER-level, result. We flag this asymmetry explicitly as a limitation: wherever the "13/14 in every arm" shorthand appears in this paper, the two claims—GBM = Holm-corrected, FM = CI-only—should be read as verbally distinct, never as an equally multiplicity-corrected result across all four configurations.

Data & Code Availability

All 14 datasets are public OpenML [15] tasks; models XGBoost [16], LightGBM [17], TabICLv2 [18], and TabDPT [19], all open-weights and token-free. Frozen result records and the code that regenerates every figure from those records accompany this draft. The development repository is `https://github.com/PeterPonyu/structured-data-fm-reliability-research`; the code is archived at Zenodo under DOI `10.5281/zenodo.21130297`, which is currently reserved on a draft deposition and becomes active when the record is published.

CRediT authorship contribution statement

Zeyu Fu: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) in order to assist with code development, experiment orchestration, and language editing. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Data availability

All 14 datasets are public OpenML tasks and the Folktables/ACS PUMS 2018 data are public US Census data; the frozen result records and figure-generation code accompany the submission, and the code is linked in the availability statement above.

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